

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2

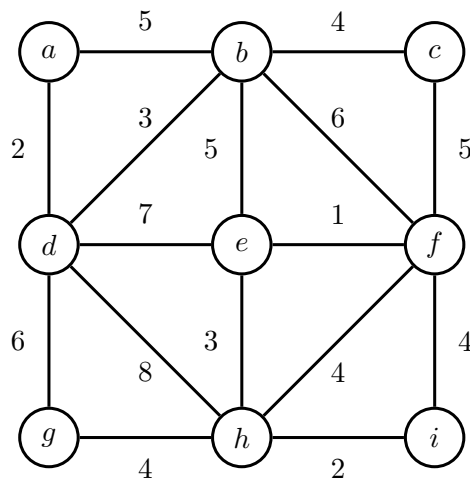
MH1301 Discrete Mathematics

Tutorial 12

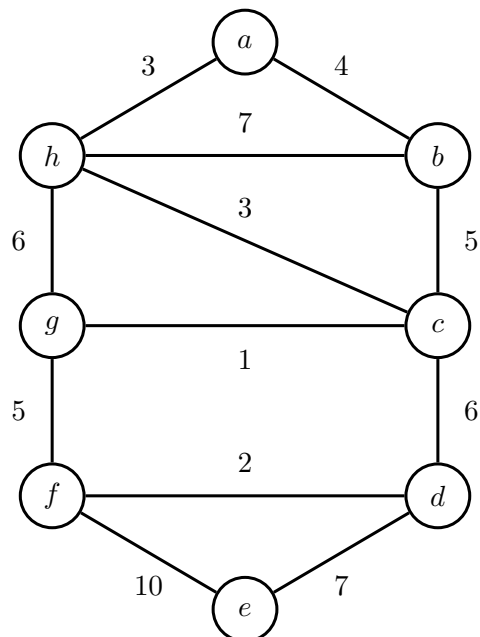
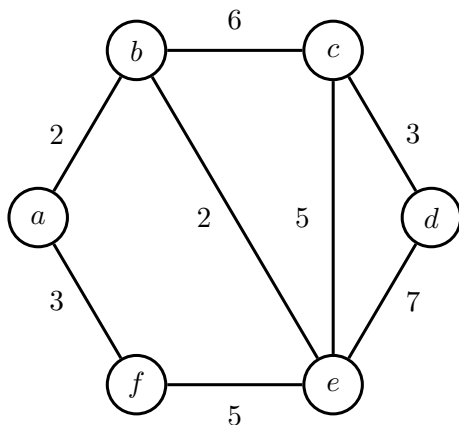
For the tutorial in Week 13 on 18 April, let us discuss the following questions.

Questions 1 and 2. (Ex.11.5.2, 11.5.4.)

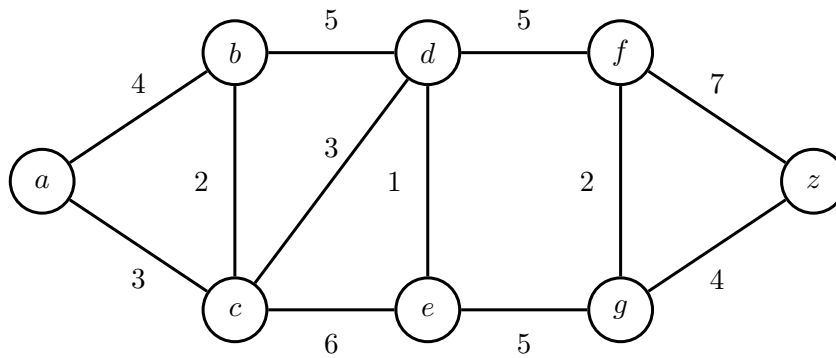
1. Use Prim's algorithm to find a minimum spanning tree for the given weighted graph.
2. Use Kruskal's algorithm to find a minimum spanning tree for the given weighted graph.



Question 3. Find a minimum spanning tree for each of the following two graphs. Use a different algorithm for each graph. For each graph, state the name of the algorithm used, and list the edges in the order that they are added to the MST.



Question 4. (Ex. 10.6.2.) Find the distance between a and z in the following graph using Dijkstra algorithm. Illustrate how the values of $d[v]$ are updated using a table.



Question 5. (Ex 10.6.4). Find the length of a shortest path between these pairs of vertices in the weighted graph in Question 4.

- (a) a and d
- (b) a and f
- (c) c and f
- (d) b and z

Questions 6 and 7. One of the basic motivations behind the Minimum Spanning Tree Problem is the goal of designing a spanning network for a set of nodes with minimum total cost. Here we explore another type of objective: designing a spanning network for which the most expensive edge is as cheap as possible.

Specifically, let $G = (V, E)$ be a connected graph with n vertices, m edges, and positive edge costs that you may assume are all distinct. Let $T = (V, E')$ be a spanning tree of G ; we define the *bottleneck edge* of T to be the edge of T with the greatest cost.

A spanning tree T of G is a *minimum bottleneck spanning tree* if there is no spanning tree T' of G with a cheaper bottleneck edge.

- 6. Is every minimum-bottleneck tree of G a minimum spanning tree of G ? Prove or give a counterexample.
- 7. Is every minimum spanning tree of G a minimum-bottleneck tree of G ? Prove or give a counterexample. (**Hint:** use the property of safe edges.)

Question 8. Let the complete graph K_n have vertices $\{1, 2, \dots, n\}$ and suppose that for each $u, v \in \{1, 2, \dots, n\}$, the edge $\{u, v\}$ has weight $w(\{u, v\}) = u + v$. For example, edge $\{1, 2\}$ will have weight $w(\{1, 2\}) = 1 + 2 = 3$, edge $\{2, 4\}$ will have weight $w(\{2, 4\}) = 2 + 4 = 6$, etc.

Determine the minimum spanning tree of this graph, and *compute* the total cost for this minimum spanning tree.

(**Hint:** apply Kruskal's algorithm to this graph, and see what are the edges added to the MST.)

Answer.

- 4) 16.
- 5) (a) 6; (b) 11; (c) 8; (d) 15.
- 6) False.
- 7) True.
- 8) $\frac{1}{2}(n+1)(n+2) - 3$.